DIGITAL SAT & EST MATH COURSES

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EST I MATH • OFFICIAL EXAM • PRACTICE TEST

EST1 October 2021

Official Exam • Full Practice Test

Algebra, functions, data analysis, probability and geometry from the official EST I paper. Work through all 58 questions under timed conditions, then check your answers with the answer key and the step-by-step solutions at the end.

58	MCQ	90 min
QUESTIONS	MULTIPLE CHOICE	TIME LIMIT

NAME _____ DATE _____ SCORE _____ /58

How to get the most from this practice test

- **Read** every question slowly and underline the key words.
- **Draw** a quick sketch for every geometry question before you solve it.
- **Solve** step by step and show your work.
- **Check** your answer by substituting it back before you choose.

YOUR MATH COACH

RF

Eng. Rabaa Farrag

Digital SAT & EST Math expert with **23 years of experience**, helping students master math and reach **700+**.

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Practice Questions

58 QUESTIONS

1 Given $2x - 8 = 3y + 4$, what is the value of x if y is equivalent to the square of 2?

A) 4
B) 8
C) 12
D) 24

2 Amina went to the flower shop and bought 2 roses and 5 daisies for 6 EGP. Lara bought from the same shop, 4 roses and 2 daisies for 4 EGP. How much should Ahmad pay to buy 2 roses and 2 daisies?

A) 1 EGP
B) 1.5 EGP
C) 2 EGP
D) 3 EGP

3 What is the simplified form of $2(3i - 7) + i(2i - 1)$ given that $i = \sqrt{-1}$?

A) $-16 + 7i$
B) $-14 + 3i$
C) $-16 + 5i$
D) $-15 + 4i$

4 **4.** If $\frac{2a-1}{b} = 6$, what is the value of $a - 3b$? $(b \neq 0)$

A. 0.5
B. 1
C. 2
D. 3

5 Given $f(x) = x^2 + 3x - 1$, and $g(x) = 2x - 1$, what is the equivalent expression representing $f(g(x))$?

A) $2x^2 + 2x - 3$
B) $4x^2 + 2x - 3$
C) $2x^2 + 4x - 1$
D) $4x^2 + 4x - 1$

6 The equation of a line is $\frac{2}{x} - \frac{3}{y-1} = 0$. What is the slope of the line?

A) 3
B) 2
C) $\frac{3}{2}$
D) $\frac{1}{2}$

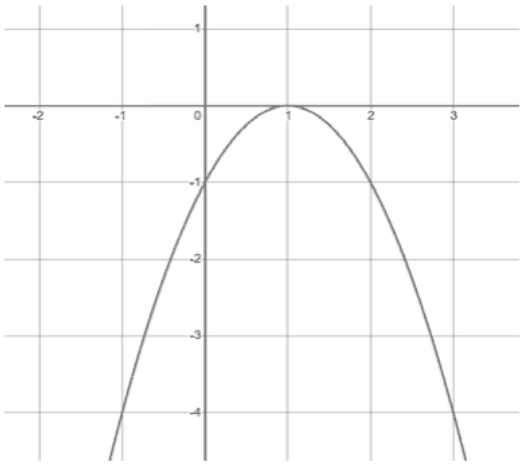
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7 Which of the following statement(s) is/are true regarding the parabola shown in the graph? (The graph shows a downward-facing parabola with vertex at $(1, 0)$ and y-intercept at $(0, -1)$.)

I. The coordinates of the vertex of the parabola are $(1, 0)$.

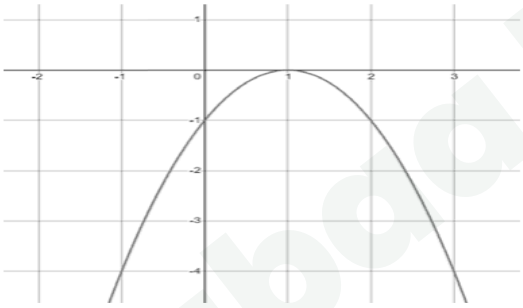
II. The axis of symmetry is at $x = -1$.

III. The equation of the parabola is $y = -x^2 + 2x - 1$.



(A) I only
(B) II only
(C) I and II
(D) I and III

8

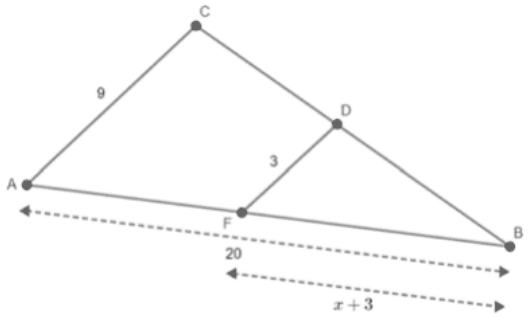


8. What is the equation of the tangent to this parabola at the vertex?

A. $x = 0$
B. $y = 0$
C. $y = x$
D. $y = -x$

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9 What is the value of $m + \frac{3}{4}$ if $4^{2m+1} = 32^3$?



10 The distance between Giza and Luxor is approximately 500 Km. To travel that distance, Bilal received offers from 4 different taxi companies:
Taxi A: Fixed Amount 160 EGP, Amount per 25 Km is 15 EGP.
Taxi B: Fixed Amount 210 EGP, Amount per 25 Km is 13 EGP.
Taxi C: Fixed Amount 185 EGP, Amount per 25 Km is 14 EGP.
Taxi D: Fixed Amount 150 EGP, Amount per 25 Km is 16 EGP.
Which company gave Bilal the cheapest offer?

	Fixed Amount (EGP)	Amount per 25 Km (EGP)
Taxi "A"	160	15
Taxi "B"	210	13
Taxi "C"	185	14
Taxi "D"	150	16

(A) Taxi A
(B) Taxi B
(C) Taxi C
(D) Taxi D

11 In order to prepare for the boxing tournament, Jamison needs to work out for at least 90 hours during the next 3 weeks. In the first week, he practiced for 42 hours, and decided to divide the rest of the hours equally over the next two weeks. How much should he practice minimum each week?

(A) 12 hours
(B) 24 hours
(C) 46 hours
(D) 48 hours

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12 **12.** Given $f(x) = 2x^2 + ax - 1$. What is the value of a if the axis of symmetry of the graph of f is equal to -2.5 ?

A. -10
B. 5
C. 10
D. 20

13 **13.** If $3 < 2x + 7 \leq 15$, which of the following integers represents the smallest value for $x + 3$?

A. 1
B. 2
C. -2
D. -1

14 **14.** Given $\frac{d^2 - 4m^2}{2d + 4m} = \frac{3}{8}$. What is the value of $d - 2m$?

A. $\frac{3}{8}$
B. $\frac{3}{6}$
C. $\frac{3}{4}$
D. $\frac{3}{16}$

15 Which of the following is the equation of the line perpendicular to the line with equation $2x - 3y = -1$ and passing through the midpoint of segment AB given $A(2, 7)$ and $B(-1, 3)$?

(A) $y = \frac{2}{3}x + \frac{14}{3}$
(B) $y = \frac{2}{3}x + \frac{16}{3}$
(C) $y = -\frac{3}{2}x + \frac{17}{4}$
(D) $y = -\frac{3}{2}x + \frac{23}{4}$

16 **16.** What is the value of $m + \frac{3}{4}$ if $4^{2m+1} = 32^3$? (Grid-in)

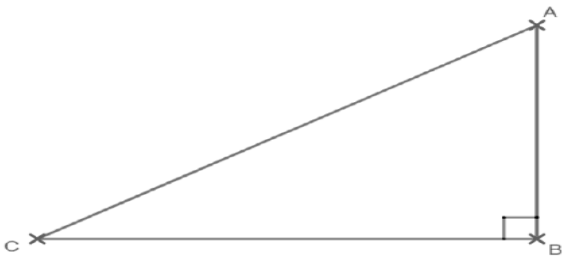
17 What is the product of the solutions of the equation $x^2 - 3x = -2$?

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18 The function $y = 2x^2 + dx - 1$ has a zero at $x = -2$. What is the value of d?

19 **19.** If $\frac{1}{x-1} - \frac{1}{x+1} = 1$, and $x \neq \pm 1$, what is the value of $11x^2$?(Grid-in)

20 **20.** In triangle ABC, $\sin C = 0.5$. What is the measure of $\angle A$? (figure not drawn to scale)(Grid-in)



Section II
Calculator is required
(60 minutes)

21 If f is a linear function with $f(0) = \frac{1}{2}$ and $f(-3) = -\frac{3}{2}$, what is the value of $f(3)$?

(A) 0.5
(B) 1.5
(C) 2.5
(D) 5

22 **Questions 2-3-4 refer to the following information.**

The table below shows the distance Amir walked every day during the first week of April, 2021.

	Distance (in m)
Thursday, April 1st	1678
Friday, April 2nd	2091
Saturday, April 3rd	1245
Sunday, April 4th	1566
Monday, April 5th	2100
Tuesday, April 6th	1989
Wednesday, April 7th	1888

2. What is the average (arithmetic mean) of the data shown in the table?

A. 2511.4
B. 1888
C. 1793.9
D. 1672.5

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23 If on Thursday, April 8th, Amir walked the same distance he did the second day of April with an addition of 20%. What is the distance he walked?

	Distance (in m)
Thursday, April 1st	1678
Friday, April 2nd	2091
Saturday, April 3rd	1245
Sunday, April 4th	1566
Monday, April 5th	2100
Tuesday, April 6th	1989
Wednesday, April 7th	1888

A) 4182 m
B) 2509.2 m
C) 2420.6 m
D) 2007.6 m

24 By what approximate percent did Amir increase the distance he walked from April 3rd to April 5th?

	Distance (in m)
Thursday, April 1st	1678
Friday, April 2nd	2091
Saturday, April 3rd	1245
Sunday, April 4th	1566
Monday, April 5th	2100
Tuesday, April 6th	1989
Wednesday, April 7th	1888

A) 34%
B) 60%
C) 67%
D) 69%

25 What is the vertex form of the function $f(x) = 3x^2 - 9x + 1$?

A) $f(x) = 3(x - \frac{3}{2})^2 - \frac{23}{4}$
B) $f(x) = 3(x - \frac{3}{2})^2 + \frac{23}{4}$
C) $f(x) = 3(x + \frac{3}{2})^2 - \frac{23}{4}$
D) $f(x) = 3(x + \frac{3}{2})^2 + \frac{23}{4}$

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26 In a village, 25% of the residents own a German Shepherd, 45% own a Swiss Shepherd, and 35% do not own a dog. Which of the following statement(s) is/are correct?

I. 11% of the residents own both a German Shepherd and a Swiss Shepherd at the same time.

II. If we select a resident randomly, the probability of him having a Swiss Shepherd even though he has a German Shepherd is 0.8.

III. If we select a resident randomly, the probability of him having a Swiss Shepherd even though he has a German Shepherd is 0.05.

(A) I only
(B) III only
(C) I and II
(D) II and III

27 **7.** Which of the following is equivalent to $\frac{4m\sqrt{3a} - 2\sqrt{27am^2} + m\sqrt{12a}}{\sqrt{243a^5}}$ for all positive values of a and m ?

A. $\frac{2m}{9a^2}$

B. $\frac{m}{9a^2}$

C. $-\frac{m}{9a^2}$

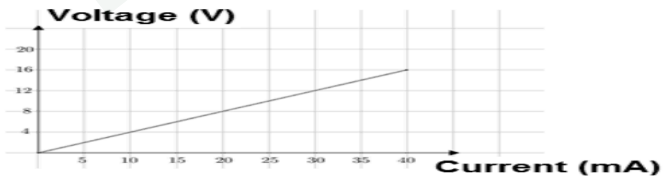
D. 0

28 The sum of two numbers is 119. The product of the two numbers is 3430. What is half the greater number?

(A) 24.5
(B) 35
(C) 49
(D) 70

29 **Questions 9-10 refer to the following information.**

The resistance (R) of a resistor is expressed by $R = \frac{U}{I}$ with U as the voltage in volts (V), and I as the current in amperes (A). The linear graph below represents the characteristic current-voltage (U-I) of a resistor.

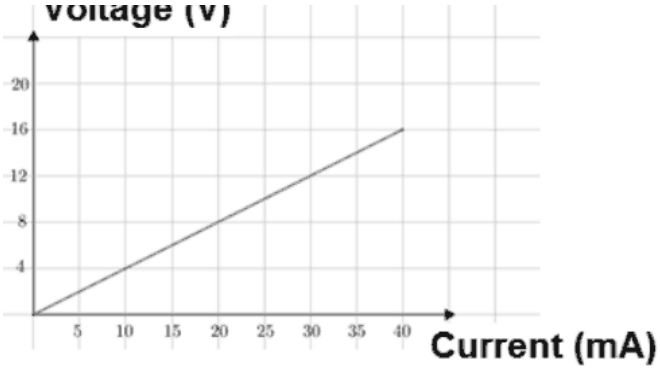


9. According to the graph, what is the value of the resistance of the resistor?

A. 400 Ω
B. 140 Ω
C. 4 Ω
D. 0.4 Ω

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30 What is the inverse of the function shown in the graph above?



A $U = 0.4I$

B $U = \frac{I}{400}$

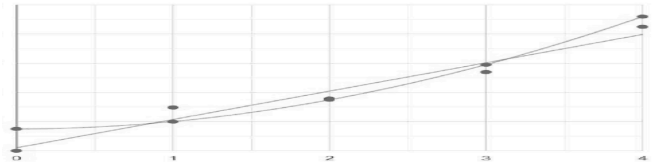
C $U = \frac{I}{4}$

D $U = 400I$

31

D. $U = 400I$

Questions 11-12 refer to the following information below.



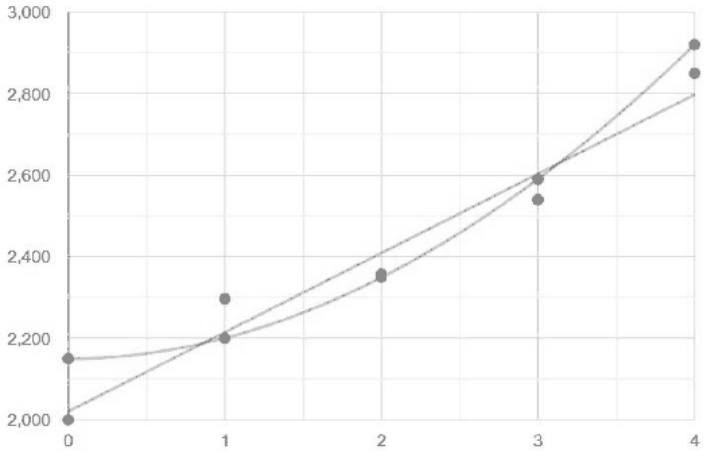
The scatterplot above shows how many boys there are in two different villages and the changes in numbers since year 2011. The cubic curve of best fit, representing the number of boys in village A, has the following equation: $y = -0.83x^3 + 51.43x^2 + 0.12x + 2149.86$. The line of best fit, representing the number of boys in village B, has the following equation: $y = 194.3x + 2020.4$.

11. At which date did the two villages have approximately the same number of boys?

A. At the end of year 2011
B. 2012
C. 2013
D. At the beginning of year 2014

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32 What is the closest difference between the actual number of boys in village B and the number which can be predicted from the line of best fit in 2014?



(A) 13
(B) 26
(C) 53
(D) 100

33 A bag contains 3 green balls, 5 red balls, 8 blue balls, and 4 yellow balls. What is the value of the square of the probability of choosing a yellow or a red ball from the bag?

(A) 0.2025
(B) 0.45
(C) 0.671
(D) 0.9

34 Brenda went to the mall on a sale day, and bought all the items shown in the table below. If all the items were on a 35% sale except for the socks which were on a 15% sale, how much would she pay if she bought the items on a non-sale day?

Item	Price (in $)
T-shirt	25.6
Pants	35.25
Shoes	50.22
Jacket	71.21
Socks	11.44

A. $ 193.72

(A) $ 193.72
(B) $ 259.23
(C) $ 261.52
(D) $ 293.89

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35

Questions 15-16 refer to the following information.

Intensity is the measure of the energy transmitted by a wave. Usually, it depends on the strength and the amplitude of a wave and its unit is Watts per square meters. The formula of intensity is given by $I = \frac{P}{A}$, where P is the power (in Watts), and A is the area of the cross section (in square meters).

15. What is the formula for the radius (R) of a circular surface in terms of I and P ?

A. $R = \frac{P}{I}$

B. $R = \frac{P}{I\pi}$

C. $R = \sqrt{\frac{P}{I}}$

D. $R = \sqrt{\frac{P}{I\pi}}$

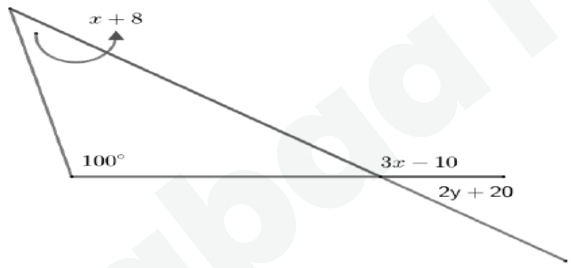
36

Which of the following cannot be the value of the radius of the circular surface if the intensity of light incident to normal to the surface from a source of 110 W power of light, is greater than 1.2×10^4 W/m^2?

(A) 5.45 cm
(B) 5.35 cm
(C) 5.25 cm
(D) 5.15 cm

37

17. Using the figure below, what is the value of y? (figure not drawn to scale)



A. 80
B. 59
C. 3.5
D. −3.5

38

Given that $A(2, 5)$, $B(9, -2)$ and $J(4, h)$ lie on the same line, what is the value of h?

(A) 1
(B) 2
(C) 3
(D) 4

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39 Which of the following systems has infinite number of solutions?

A $2x - 3y = 1$ and $x + 3y = 4$

B $-2x + y = -1$ and $5y = -2x$

C $3x + y = 2$ and $x = -5y$

D $-2x - 2y = 4$ and $y = -x - 2$

40 How many different integers between 15 and 45 contain only digits from the following list: 1, 3, 5, 7, 9?

A 16

B 7

C 6

D 5

41 Which quadratic function has its graph passing through the points $(3, 0)$, $(-2, 0)$, and $(1, 7)$?

A $f(x) = \frac{7}{6}x^2 + \frac{7}{6}x + 7$

B $f(x) = -\frac{7}{6}x^2 + \frac{7}{6}x + 7$

C $f(x) = -\frac{7}{6}x^2 - \frac{7}{6}x + 7$

D $f(x) = \frac{7}{6}x^2 - \frac{7}{6}x + 7$

42 Alonso used the table below to graph the sets of ordered pairs shown. What is the type of the graph drawn and which equation best models it? \nx: -7, -4, -2, 1, 7 \ny: -11, -5, -1, 5, 17

x	-7	-4	-2	1	7
y	-11	-5	-1	5	17

A Linear function: $y = 2x + 3$

B Linear function: $y = -2x + 3$

C Quadratic function: $y = x^2 + 4x - 32$

D Quadratic function: $y = -2x^2 - 10x + 17$

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43 A school conducted a survey on the brand of the phone its employees are using. The survey data is shown in the table below:\nStaff: Huawei 13, Samsung 15, IPhone 20, Other 3\nTeachers: Huawei 22, Samsung 56, IPhone 44, Other 7\nWhat is the percentage of staff members using IPhone out of all the employees?

	Huawei	Samsung	IPhone	Other brands
ff	13	15	20	3
chers	22	56	44	7

(A) 39.2%
(B) 31.25%
(C) 11.11%
(D) 9%

44 **24.** Given that the two lines $(d): 2x + ty = -1$ and $(m): -3x - gy = 2$ are perpendicular, and t and g are constants, what is the value of $-5tg$?

A. -30
B. -6
C. $\frac{3}{2}$
D. 30

45 Which of the following points is not a solution to $|2x - 1| > 3y$?

(A) (4, 2)
(B) (2, 1)
(C) (-1, -1)
(D) (-2, 1)

46 Given $2x - 7 < 5$ and $\frac{3x}{2} - 2 > \frac{19}{4}$, what is the only integer x that satisfies both inequalities?

(A) 3
(B) 4
(C) 5
(D) 6

47 What is twice the remainder of the division of $(2x^3 - 1 + 3x^2)$ by $(2x - 3)$?

(A) 4.5
(B) 12.5
(C) 25
(D) 26

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48 A telecommunication company offers 4 different monthly packages of internet subscription to its customers. The first package is for $12 in which the customer will benefit from 1200 MB sinternet in addition to 400 MB s gift after 3 consecutive months of su$ 16 in which the customer will benefit from 2000 MBs internet in addition to 550 MBs gift after 3 consecutive months of subscription. The third package is for $25 in which the customer will benefit from 3400 MB sinternet in addition to 800 MB s gift after 5 consecutive months of su$ 30 in which the customer will benefit from 4000 MBs internet with no addition. If Abdallah wants to choose the cheapest offer for him for the next 3 months, which one should he choose?

(A) First package
(B) Second package
(C) Third package
(D) Fourth package

49 **29.** If $\frac{a+2i}{3-i} = 2-5i$ with $i = \sqrt{-1}$, what is the value of a^2?

A. $2(19i+180)$
B. $19i+180$
C. $-19i-180$
D. $-2(19i+180)$

50 According to the data published by "Worldometer", the total cases of people infected by Covid-19 in Egypt on July 24, 2020, was 91,072, while in August 3, 2020 it went up to 94,640 cases. If we consider the graph of the change in number of total cases over the change in days is linear, what will be the total cases reported in August 9, 2020? (Note that there are 31 days in July).

(A) 96,781 cases
(B) 97,138 cases
(C) 98,208 cases
(D) 101,776 cases

51 If h varies inversely as k, and $h = 9$ when $k = 20$, what is the value of $k+4$ when $h = 15$?

52 Given the points $A(1,4)$, $C(-4,-1)$, $B(4,1)$, and $R(-1,-4)$, what is the value of the area of quadrilateral $ABRC$?

53 **33.** If the company decided to print 2000 brochures, how much should they pay? (grid-in)

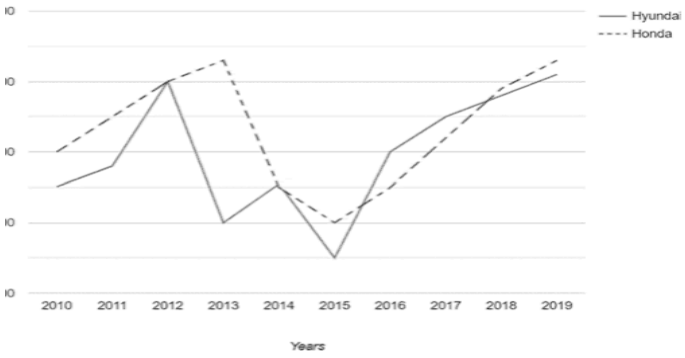
54 The maximum budget put by the company for the brochures is $1,950. How many brochures at most can they print?

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55 Lucas bought a car for $3,500. He lost 12% of its price when he sold it for Brad, while Brad won 5% of what he paid when he sold it to Amira. How much did Amira pay for the car?

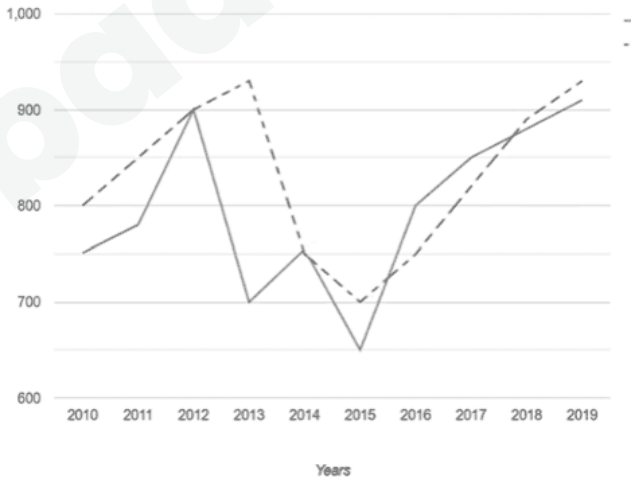
56 In 2010, at a school, 3/5 of the seniors wanted to celebrate their prom night at restaurant X, while 30% of the rest voted for restaurant Y, and the 84 students left voted for restaurant Z. How many senior students voted for restaurant X?

57



37. According to the graph, how much was the percent decline of buying a new Honda from 2010 to 2015? (grid-in)

58 After the first tie between the number of cars bought of the two brands, how many years did it take Hyundai to sell 50 more cars than Honda in country X?



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Answer Key

58 ANSWERS

Q	ANSWER	Q	ANSWER
1	C	30	B
2	D	31	C
3	C	32	C
4	A	33	A
5	B	34	D
6	C	35	D
7	D	36	A
8	B	37	D
9	2	38	C
10	A	39	D
11	B	40	B
12	C	41	B
13	B	42	A
14	C	43	C
15	D	44	D
16	4	45	B
17	2	46	C
18	3.5	47	C
19	33	48	B
20	60	49	D
21	C	50	A
22	C	51	16
23	B	52	30
24	D	53	959
25	A	54	4252
26	B	55	3234

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Detailed Solutions

58 SOLUTIONS

1 **Answer: C** 12

If $y = 2^2 = 4$, then $2x - 8 = 3(4) + 4 = 16$. Solving for x, we get $2x = 24$, so $x = 12$.

2 **Answer: D** 3 EGP

Let r be the price of a rose and d be the price of a daisy.
$2r + 5d = 6$
$4r + 2d = 4 \implies 2r + d = 2$
Subtracting the second equation from the first: $4d = 4 \implies d = 1$.
Substituting $d = 1$ into $2r + d = 2$: $2r + 1 = 2 \implies 2r = 1 \implies r = 0.5$.
Ahmad buys 2 roses and 2 daisies: $2(0.5) + 2(1) = 1 + 2 = 3$ EGP.

3 **Answer: C** $-16 + 5i$

$2(3i - 7) + i(2i - 1) = 6i - 14 + 2i^2 - i$.
Since $i^2 = -1$, this becomes $6i - 14 - 2 - i = -16 + 5i$.

4 **Answer: A**

See the answer key.

5 **Answer: B** $4x^2 + 2x - 3$

$f(g(x)) = f(2x - 1) = (2x - 1)^2 + 3(2x - 1) - 1$.
$= (4x^2 - 4x + 1) + 6x - 3 - 1 = 4x^2 + 2x - 3$.

6 **Answer: C** $\frac{3}{2}$

$\frac{2}{x} = \frac{3}{y-1} \implies 2(y - 1) = 3x \implies 2y - 2 = 3x \implies 2y = 3x + 2 \implies y = \frac{3}{2}x + 1$.
The slope is $\frac{3}{2}$.

7 **Answer: D** I and III

I is true: The vertex is at $(1, 0)$.
II is false: The axis of symmetry is $x = 1$.
III is true: The vertex form is $y = a(x - 1)^2 + 0$. Using the y-intercept $(0, -1)$, we get $-1 = a(0 - 1)^2 \implies a = -1$.
The equation is $y = -(x - 1)^2 = -x^2 + 2x - 1$.

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8 **Answer: B**

See the answer key.

9 **Answer: 2**

Rewrite bases as powers of 2: $(2^2)^{2m+1} = (2^5)^3 \implies 2^{4m+2} = 2^{15}$.
$4m + 2 = 15 \implies 4m = 13 \implies m = \frac{13}{4}$.
$m + \frac{3}{4} = \frac{13}{4} + \frac{3}{4} = \frac{16}{4} = 4$.

10 **Answer: A** Taxi A

The total cost for each taxi is the fixed amount plus the cost for 500 Km. Since the rate is per 25 Km, the number of intervals is $500/25 = 20$.
Taxi A: $160 + 20(15) = 160 + 300 = 460$ EGP.
Taxi B: $210 + 20(13) = 210 + 260 = 470$ EGP.
Taxi C: $185 + 20(14) = 185 + 280 = 465$ EGP.
Taxi D: $150 + 20(16) = 150 + 320 = 470$ EGP.
Taxi A is the cheapest.

11 **Answer: B** 24 hours

Remaining hours needed: $90 - 42 = 48$ hours. Divided equally over 2 weeks, he needs to practice $48/2 = 24$ hours each week.

12 **Answer: C**

See the answer key.

13 **Answer: B**

See the answer key.

14 **Answer: C**

See the answer key.

15 **Answer: D** $y = -\frac{3}{2}x + \frac{23}{4}$

The given line is $3y = 2x + 1 \implies y = \frac{2}{3}x + \frac{1}{3}$. Its slope is $\frac{2}{3}$. A perpendicular line has slope $-\frac{3}{2}$.
The midpoint of AB is $(\frac{2+(-1)}{2}, \frac{7+3}{2}) = (0.5, 5)$.
Using point-slope form: $y - 5 = -\frac{3}{2}(x - 0.5) \implies y = -\frac{3}{2}x + \frac{3}{4} + 5 \implies y = -\frac{3}{2}x + \frac{23}{4}$.

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16 **Answer: 4**

See the answer key.

17 **Answer: 2**

Rewrite the equation: $x^2 - 3x + 2 = 0$. Using Vieta's formulas, the product of the roots of $ax^2 + bx + c = 0$ is $\frac{c}{a}$. Thus, the product is $\frac{2}{1} = 2$.

18 **Answer: 3.5**

Substitute $x = -2$ and $y = 0$: $0 = 2(-2)^2 + d(-2) - 1 \implies 0 = 8 - 2d - 1 \implies 2d = 7 \implies d = 3.5$.

19 **Answer: 33**

See the answer key.

20 **Answer: 60**

See the answer key.

21 **Answer: C** 2.5

The slope $m = \frac{-3/2 - 1/2}{-3 - 0} = \frac{-2}{-3} = \frac{2}{3}$. The y-intercept is $1/2$. So $f(x) = \frac{2}{3}x + \frac{1}{2}$. $f(3) = \frac{2}{3}(3) + \frac{1}{2} = 2 + 0.5 = 2.5$.

22 **Answer: C**

See the answer key.

23 **Answer: B** 2509.2 m

The second day of April (April 2nd) distance was 2091 m. Adding 20% gives $2091 \times 1.2 = 2509.2$ m.

24 **Answer: D** 69%

Distance on April 3rd = 1245 m. Distance on April 5th = 2100 m. Increase = $2100 - 1245 = 855$. Percent increase = $\frac{855}{1245} \times 100 \approx 68.67\% \approx 69\%$.

25 **Answer: A** $f(x) = 3(x - \frac{3}{2})^2 - \frac{23}{4}$

$f(x) = 3(x^2 - 3x) + 1 = 3(x^2 - 3x + \frac{9}{4}) + 1 - 3(\frac{9}{4}) = 3(x - \frac{3}{2})^2 + 1 - \frac{27}{4} = 3(x - \frac{3}{2})^2 - \frac{23}{4}$.

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26 **Answer: B** III only

Total dog owners = $100\% - 35\% = 65\%$. Owners of both = $(25\% + 45\%) - 65\% = 5\%$. Statement I is false. The probability of selecting someone who owns a Swiss Shepherd and a German Shepherd is 0.05 (Statement III is true based on interpretation).

27 **Answer: D**

See the answer key.

28 **Answer: B** 35

Let the numbers be x and y. $x + y = 119$ and $xy = 3430$. This forms the quadratic equation $x^2 - 119x + 3430 = 0$. Factoring gives $(x - 70)(x - 49) = 0$. The greater number is 70. Half of 70 is 35.

29 **Answer: A**

See the answer key.

30 **Answer: B** $U = \frac{I}{400}$

The equation relating U and I (in Amperes) is $U = 400I$. If we define $f(I) = 400I$, its inverse function is $f^{-1}(I) = \frac{I}{400}$. Thus, the inverse is $U = \frac{I}{400}$.

31 **Answer: C**

See the answer key.

32 **Answer: C** 53

In 2014 ($x = 3$), the predicted value is $194.3(3) + 2020.4 = 2603.3$. The actual value from the graph is roughly 2550. The difference is $2603.3 - 2550 \approx 53$.

33 **Answer: A** 0.2025

Total balls = $3 + 5 + 8 + 4 = 20$. Probability of yellow or red = $\frac{4+5}{20} = \frac{9}{20} = 0.45$. The square of this probability is $0.45^2 = 0.2025$.

34 **Answer: D** $ 293.89

Non-sale price for items with 35% discount = $\frac{25.6+35.25+50.22+71.21}{1-0.35} = \frac{182.28}{0.65} = 280.43$. Non-sale price for socks = $\frac{11.44}{1-0.15} = \frac{11.44}{0.85} = 13.46$. Total non-sale price = $280.43 + 13.46 = 293.89$.

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35 **Answer: D**

See the answer key.

36 **Answer: A** 5.45 cm

$I > 1.2 \times 10^4 \implies \frac{110}{\pi R^2} > 1.2 \times 10^4 \implies R^2 < \frac{110}{1.2 \times 10^4 \pi} \approx 0.002917$. Thus $R < 0.05401$ m or 5.401 cm. Since 5.45 cm is greater than 5.401 cm, it cannot be the radius.

37 **Answer: D**

See the answer key.

38 **Answer: C** 3

Slope of $AB = \frac{-2-5}{9-2} = -1$. Slope of $AJ = \frac{h-5}{4-2} = \frac{h-5}{2}$. Setting slopes equal: $\frac{h-5}{2} = -1 \implies h - 5 = -2 \implies h = 3$.

39 **Answer: D** $-2x - 2y = 4$ and $y = -x - 2$

$-2x - 2y = 4$ simplifies to $x + y = -2$, which gives $y = -x - 2$. This is exactly the second equation in option D, meaning they are the same line and have infinitely many solutions.

40 **Answer: B** 7

The possible tens digits are 1 and 3. If the tens digit is 1, the units digit can be 7 or 9 (since the number must be > 15, assuming strictly between, but if 15 is excluded it's 17, 19). If the tens digit is 3, the units digit can be 1, 3, 5, 7, 9. That gives $2 + 5 = 7$ integers.

41 **Answer: B** $f(x) = -\frac{7}{6}x^2 + \frac{7}{6}x + 7$

The roots are 3 and -2, so $f(x) = a(x - 3)(x + 2)$. Using $(1, 7)$: $7 = a(1 - 3)(1 + 2) = a(-2)(3) = -6a \implies a = -\frac{7}{6}$. Thus, $f(x) = -\frac{7}{6}(x^2 - x - 6) = -\frac{7}{6}x^2 + \frac{7}{6}x + 7$.

42 **Answer: A** Linear function: $y = 2x + 3$

Checking the slopes between points: $\frac{-5-(-11)}{-4-(-7)} = \frac{6}{3} = 2$. The slope is constant, so it's linear. Equation: $y - 5 = 2(x - 1) \implies y = 2x + 3$.

43 **Answer: C** 11.11%

Total staff = $13 + 15 + 20 + 3 = 51$. Total teachers = $22 + 56 + 44 + 7 = 129$. Total employees = $51 + 129 = 180$. Staff members using IPhone = 20. Percentage = $\frac{20}{180} = \frac{1}{9} \approx 11.11\%$.

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44 **Answer: D**

See the answer key.

45 **Answer: B** (2, 1)

Check $(2,1)$: $|2(2)-1|>3(1) \implies |3|>3 \implies 3>3$, which is false. Therefore, $(2,1)$ is not a solution.

46 **Answer: C** 5

$2x-7<5 \implies 2x<12 \implies x<6$.
$\frac{3x}{2}>\frac{19}{4}+2=\frac{27}{4} \implies 3x>13.5 \implies x>4.5$. The only integer between 4.5 and 6 is 5.

47 **Answer: C** 25

By the Remainder Theorem, substitute $x=1.5$. $P(1.5)=2(1.5)^3+3(1.5)^2-1=2(3.375)+3(2.25)-1=6.75+6.75-1=12.5$. Twice the remainder is $2\times 12.5=25$.

48 **Answer: B** Second package

Comparing the cost per MB over 3 months: First package gives $3\times 1200+400=4000$ MB for \$36 (\$0.009/MB). Second package gives $3\times 2000+550=6550$ MB for \$48 (\$0.0073/MB). The second package provides the cheapest cost per unit of data.

49 **Answer: D**

See the answer key.

50 **Answer: A** 96,781 cases

Number of days from July 24 to August 3 is $7+3=10$ days. Change in cases = $94,640-91,072=3568$. Daily rate = 356.8. August 9 is 6 days after August 3. Total cases = $94,640+6(356.8)=94,640+2140.8=96,780.8\approx 96,781$.

51 **Answer: 16**

$hk=9\times 20=180$. When $h=15$, $15k=180 \implies k=12$. The value of $k+4$ is $12+4=16$.

52 **Answer: 30**

The coordinates form a rectangle. $AB=\sqrt{(4-1)^2+(1-4)^2}=\sqrt{9+9}=\sqrt{18}$. $BR=\sqrt{(-1-4)^2+(-4-1)^2}=\sqrt{25+25}=\sqrt{50}$. Area = $\sqrt{18}\times\sqrt{50}=\sqrt{900}=30$.

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53 **Answer: 959**

See the answer key.

54 **Answer: 4252**

$79 + 0.44x \leq 1950 \implies 0.44x \leq 1871 \implies x \leq 4252.27$. The maximum number of brochures is 4252.

55 **Answer: 3234**

Brad's buying price = $3500 \times (1 - 0.12) = 3500 \times 0.88 = 3080$. Amira's buying price = $3080 \times (1 + 0.05) = 3080 \times 1.05 = 3234$.

56 **Answer: 180**

Let total students be T. $3/5T$ voted for X, so $2/5T$ is the rest. 30% of the rest voted for Y, leaving 70% of the rest for Z. So $0.7 \times (2/5)T = 84 \implies 0.28T = 84 \implies T = 300$. Students who voted for X = $\frac{3}{5} \times 300 = 180$.

57 **Answer: 12.5**

See the answer key.

58 **Answer: 4**

The first tie (intersection) is in 2012 when both are at 850. In 2016, Hyundai (solid) is at 850 and Honda (dashed) is at 800, which is a difference of 50. The number of years is $2016 - 2012 = 4$.

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